The effect of three hot drinks, coffee, warm milk, and tea, on salivary cortisol levels.

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Introduction

"Saliva cortisol is a steroid hormone that is produced in the hypothalamic-pituitary-adrenal axis and secreted into saliva when persons are under stress" [1]. Young people would be experiencing more mental health problems, anxiety and depression, and the age group of 18 - 34 has a relatively high occupation of stress [2]. which means that young people would consume hot drinks to help them to reduce stress. "*Coffee and tea are the two most frequently consumed hot drinks across the world, with other widely consumed hot drinks such as warm milk, and hot chocolate"* [3]. It was found that the assumption of hot drinks would affect the saliva cortisol level. Factors that would change the saliva cortisol level such as daily rhythm, caffeine ingestion, recent infection, and antibiotic intake [4]. So, we believe that taking coffee, tea and warm milk will resulting different saliva cortisol levels, and we wish to investigate whether these different types of hot drinks have different effects on salivary cortisol. This study investigates the difference in salivary cortisol levels after drinking three hot drinks: coffee, warm milk, and tea by conducting a designed experiment. The null hypothesis is that there is no difference in salivary cortisol levels between coffee, warm milk, and tea after the drink. The alternative hypothesis is that the selection of three hot drinks does have a difference in salivary cortisol levels.

Methods

To investigate the different effectiveness of hot drinks on salivary cortisol levels, a designed experiment study was conducted with 60 subjects who were in aged 18 – 34. Subjects were randomly selected from the city of Arcadia in the Island by using R.

Subjects were randomly receiving 250ml of one of the three types of hot drinks: coffee, tea, and warm milk. Randomization was done by python which ensured subjects were equally likely to receive a drink. Coffee and tea have strong effectiveness after 30 minutes [5][6]. So, the salivary cortisol level was collected via salivary test 30 minutes after the subjects had consumed the drinks.

A total of 57 samples were collected where 19 samples for each hot drink. Data was collected into a table and analysed by using R.

The study aims to investigate the different effects of hot drinks. This study focuses on finding variance between the response variables - salivary cortisol levels and explanatory variables - types of drink. Since the explanatory variables are independent and all belong to a category. A one-way ANOVA model is useful for analysing the data and investigating the difference.

The response variable – salivary cortisol level is denoted as Y_{ik}, and the error is denoted as ε_{ik}, where i is the level of the response variable, and k is the independent replication of each level i. Let μ_1 be the mean effect of the reference level of level i, and α_i be the increment effect to level i.

Three assumptions were made to use the one-way ANOVA model: The response variables from all drinks are normally distributed, each group of data has an equal variance, and variables in the sample are independent. Without these assumptions, the results are not reliable, and the ANOVA model is not validated.

The one-way ANOVA model with salivary cortisol level as response variable and different types of hot drinks as explanatory variables is given as follows:

$$Y_{ik} = \mu_1 + \alpha_i + \varepsilon_{ik}$$

Where $\varepsilon_1, \dots, \varepsilon_{57} \sim iid\ N(0,\ \sigma^2)$

The null hypothesis H_0 is:

$$H_0 :\ \mu_1 = \mu_2 = \mu_3$$

And the alternative hypothesis H_1 is:

$$H_1 :\ \mu_1 \neq \mu_2 \neq \mu_3$$

Results

The mean of each group of drinks is shown in Table 1. It is clear that there is a difference between the mean salivary cortisol level effect by coffee (0.3020000), milk (0.2261053) and tea (0.2707368). Visually, $\mu_1 \neq \mu_2 \neq \mu_3$. Figure 1 is the box plot of the data, it visually shows the spread of response variables within each factor. The box plot also shows that the salivary cortisol levels of each hot drink is different.

By running the summary in R, the F-distribution under H_0 is 9.898, and the p-value ($p < 0.000218$).

Figure 2 is a Q-Q plot generated from R. The random residuals in the Q-Q plot are approximately in a linear model. It shows that the response variables are normal distributed approximately, which the assumptions of normality are proved.

One of the assumptions made by the ANOVA model is equal variance. However, the length of the box plot of each factor in Figure 1 is not equal. But since the sample size of the three hot drinks is the same, the ratio of the lowest sd (0.0314429) and the highest sd (0.0611010) is smaller than 2. Thus, the assumption is not violated due to the rule of thumb. With the three assumptions held, the one-way ANOVA model is approximately validated.

Treatment	**Coffee**	**Milk**	**Tea**
Mean (µg/dL)	0.3020000	0.2261053	0.2707368
sd	0.0604584	0.0314429	0.0611010

Table 1: Table of means and sd of three hot drinks.

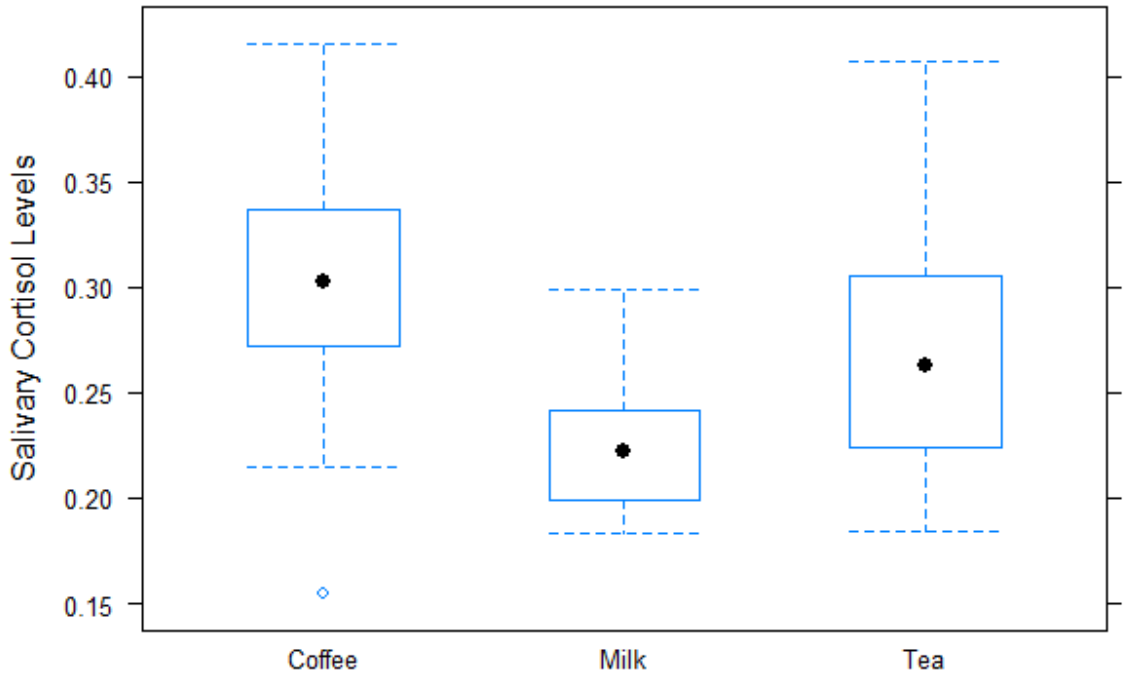

Figure 1. Box plot of salivary cortisol level measured in different hot drinks.

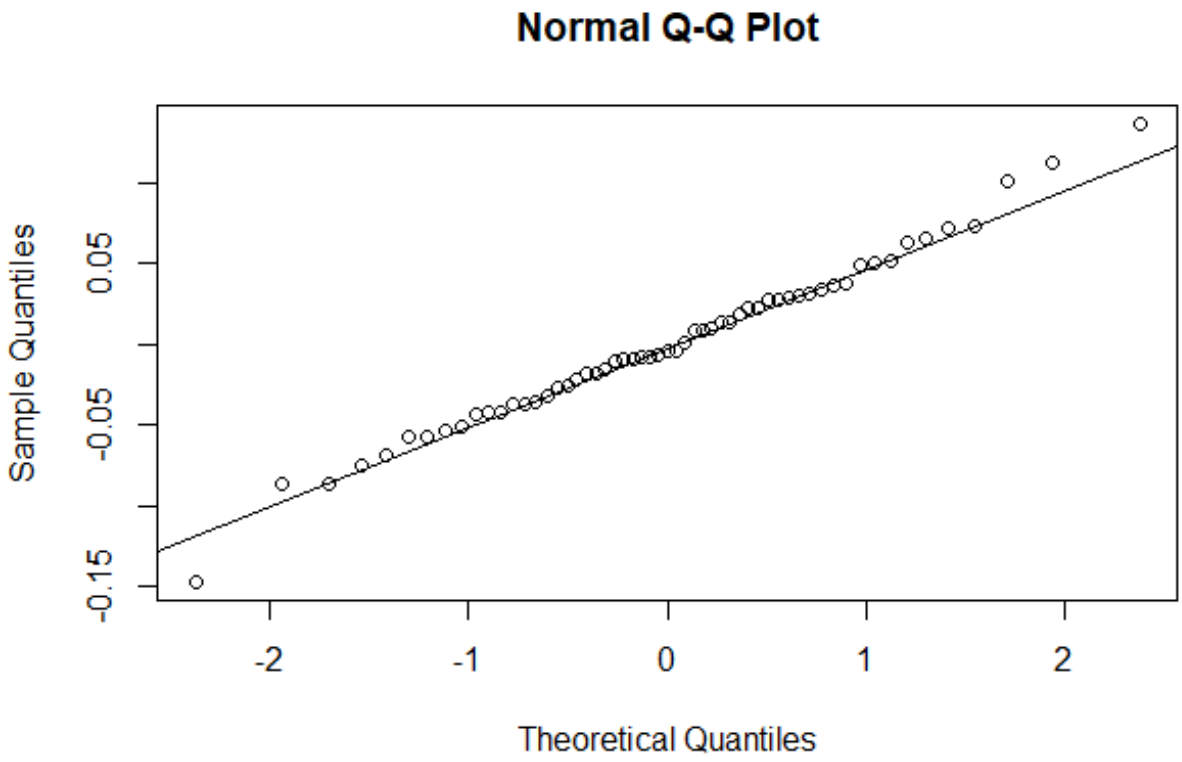

Figure 2. Quantile-Quantile plot that validates the assumption of normal distribution.

Discussion and Conclusion

With the one-way ANOVA model analysis, the p-value ($p < 0.000218$) suggested that there is strong evidence to reject the null hypothesis. Therefore, different types of hot drinks have a different effect on salivary cortisol levels. However, the study has a few limits. A mistake was made during the random sample collection. It is recommended that we check on whether the subject can consume hot drinks even if they accept the study before collecting data, this causes a reduced sample size which will potentially affect the results.

The equal variance assumptions of the ANOVA model might violate due to the relatively high difference between standard deviations. The difference is approximate two times. Without equal variance, the result from the analysis is not reliable.

In summary, by conducting a controlled experiment of 57 subjects, and analysing the results with the one-way ANOVA model, the study has found that different types of hot drinks affect salivary cortisol levels. Which answers the research question that different types of hot drinks could potentially affect mental stress, which is measured by salivary cortisol levels.

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